Prof. Erol Gelenbe Spring 2017

Assignment 1
Analysis and simulation of a multiple access system
Due Tuesday 31st January 2017

1) Discuss in rigorous mathematical terms the stability problem for the random access broadcast channel (slotted ALOHA). Derive the conditions for network stability in terms of the number of blocked packets, and for the retransmission control that maximises the channel's throughput. You should review several papers on the subject ranging from the 1970's to recent years.

2) Write a discrete event simulation program for the slotted ALOHA system with the assumption that the number of "fresh" packet transmissions follows a Poisson distribution in each slot, and that each blocked terminal retransmits independently with probability p in each successive slot. Run simulations for several thousand time slots in each case that show:

   (a) the number of blocked packets,

   (b) the channel throughput.

   Use different values of the "fresh" packet arrival rate λ and of p for different simulation runs. Also, simulate the channel with an external arrival rate that can lead to stability, together with the value of p which should lead to the maximum channel throughput.

Prof Gelenbe, Spring 2017, Assignment 2

Performance of a Network Router

Return by: 21/02/2017

In this assignment we are going to investigate the performance of a single router, as part of a communications system. Packets arrive at the router through its data lines. Each packet is processed by the router in order to determine its next destination, and it is then forwarded accordingly.

In order to study the performance of the router, you should make the following assumptions:

- The packets arrive at the router following a Poisson process with an average of λ packets / second.
- The service time of the router is an exponentially distributed random variable, with an average of $1/\mu$ seconds.
- The router is able to process only one packet at a time. If a packet arrives while the router is busy processing another packet, it enters the router's buffer and waits there until it is its turn to be processed in FIFO order.
- There is no limitation for the size of the queue.

Question 1: Design and implement a random number generator for an exponential distribution.

Question 2: Design and implement a discrete event simulation program which will be used to evaluate the performance of the router.

Question 3: Use your simulation program to evaluate the performance of the router under various incoming traffic rates. More specifically:

- The average service time of the router is: $1/\mu = 1$ milliseconds.
- We want to evaluate its performance for nine different average incoming traffic rates: $\lambda = [100, 200, 300, 400, 500, 600, 700, 800, 900]$ packets / second.
- The duration of the simulation will be 10 seconds.

You should produce:

a) A graph of the average router utilization vs. the nine traffic rates
b) A graph of the average waiting time of a packet in the queue vs. the nine traffic rates
c) A graph of the average queue length vs. the nine traffic rates

Note: Each graph will consist of nine points, each point depicting the average value of the respective parameter (e.g. router utilization) for a specific traffic rate. Instead of using only one simulation run for calculating this average value, you should use multiple runs and take the average over all of them. A suggested value for the number of runs is 50.

Prof Erol Gelenbe, Coursework 3, Spring 2017

Analysis of Coupled Systems

To be returned on 7th March 2017

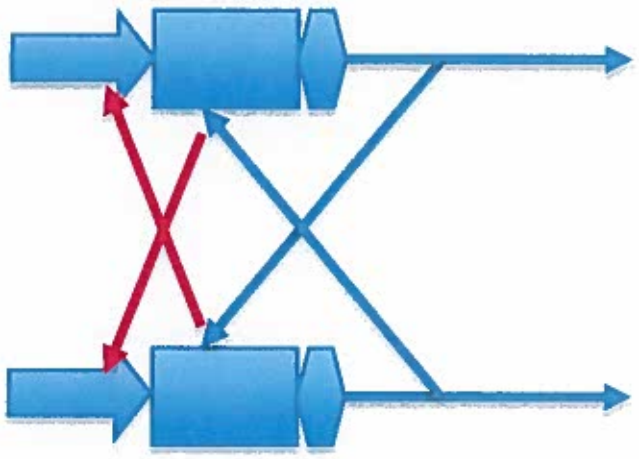
Two queues received normal (positive) customer arrivals from the outside world at Poisson processes of rates λ_1 and λ_2, and have independent and identically and exponentially distributed service times with service rates, respectively μ_1 and μ_2.

With probability p_1, the customer leaving queue 1, goes to queue 2 and moves a customer, from queue 2 to queue 1, if that customer exists (i.e. if queue 2 has at least one customer), otherwise only queue 1 is decremented and nothing happens. With probability $(1-p_1)$ the customer leaving queue 1 just departs from the system.

Similarly with probability p_2, the customer leaving queue 2, goes to queue 1 and moves a customer, from queue 1 to queue 2, if that customer exists (i.e. if queue 1 has at least one customer), otherwise only queue 2 is decremented and nothing else happens. With probability $(1-p_2)$ the customer leaving queue 2 just departs from the system.

Derive the equations in steady state for $p(n,m)$ for $n\geq 0, m\geq 0$, the joint probability distribution of the queue length at both queues.

Obtain and prove the steady-state distribution of queue length.



Distributed Computation and Networks, Spring 2017

Prof Erol Gelenbe

Assignment 4, Performance Evaluation of Cloud Servers

Due 28th March, 2017

A Cloud Service Provider (CSP) has N servers, which can each be modelled by a single server with FIFO service and exponentially distributed service times of rates r(1), ... , r(N) where $r(i)^{-1}$ is measured in seconds. We will assume that a Job Dispatcher operates within the CSP to send jobs that arrive, onwards to one of the servers, and each server has its own input queue.

The CSP receives jobs at a rate of a (jobs/sec), and will send a proportion p(i) of jobs to server i.

We will use n(i) to represent the number of jobs that are present at server I in steady-state.

1) What is the joint probability distribution in steady-state P(n(1), ... , n(N)) ?
2) What is the average response time W of jobs arriving to the system?
3) How should we choose the p(i) if we wish to minimise W?
4) Assuming the optimum assignment probabilities that you will find in 3), what is the value of the average number of jobs J(i) in server i ?
5) Given your result in 3), how would you build a practical heuristic rule for dispatching jobs to the servers so that the optimum value W* of W may be attained?
6) So far we have assumed that each server has its own queue. Would it be better to have one single common queue at the entrance to the system, before the Job Dispatcher, and then for the Job Dispatcher to send jobs to servers as soon as they are free?